MOHAMED KHIDER UNIVERSITY OF BISKRA

FACULTY OF ARCHITECTURE, CIVIL ENGINEERING, HYDRAULICS

DEPARTMENT OF CIVIL ENGINEERING AND HYDRAULICS

LABORATORY REPORT N°2

DIRECT SHEAR TEST

Determination of Soil Strength Parameters

(Dense Sand Sample)

Student

Oussama ACHOURI

Supervisor

Dr. Feia SADOK

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1. INTRODUCTION

TEST OVERVIEW

The direct shear test is a fundamental experiment in geotechnical engineering used to determine the shear strength of soil. The results provided in this report analyze both the **Peak Strength** (at failure) and the **Permanent Strength** (at large displacements) for a sample of dense sand.

1.1 Objective of the Test

The primary objective is to determine the shear strength parameters of the **dense sand** sample:

- **Cohesion** (c) - The shear strength at zero normal stress.
- **Angle of internal friction** (ϕ) - The rate of increase in shear strength with normal stress (both Peak and Permanent).

1.2 Equipment Used

Direct Shear Apparatus:	Complete testing frame with shear box
Shear Box:	Area $A = 3600$ mm^2 (60mm $\times$ 60mm)
Loading System:	Vertical loading platform and weights
Measurement:	Dynamometric ring (force measurement)
Displacement:	Dial gauge (0.01 mm precision)

2. THEORETICAL BACKGROUND

The shear strength of the soil is assumed to follow the **Mohr-Coulomb failure criterion**. This theory suggests that failure occurs when the shear stress (τ) on a failure plane reaches a critical value, defined by:

$$\tau_f = c + \sigma_n \tan(\phi) \tag{1}$$

Where:

τ_f : Shear strength (kPa)

c : Cohesion intercept (kPa)

σ_n : Normal stress on the failure plane (kPa)

ϕ : Angle of internal friction (degrees)

For **dense sand**, we expect a peak strength due to the interlocking of grains, followed by a reduction to a residual (constant volume) strength as the soil dilates (expands in volume) during shear.

3. OPERATING PROCEDURE

Step 1. **Preparation:** Loosen the screws of the dynamometric ring.

Step 2. **Assembly:** Assemble the two half-boxes, then place the base, a draining stone, and a grooved plate.

Step 3. **Sample Preparation:** Collect dense sand sample (m = 180g) and place in the shear box.

Step 4. **Compaction:** Lightly compact with tamper to achieve Density Index $I_d = 0.9$.

Step 5. **Completion:** Add upper grooved plate, porous stone, and lid.

Step 6. **Mounting:** Place box in frame, connect upper half-box to dynamometric ring.

Step 7. **Alignment:** Align vertical loading system and ensure contact.

Step 8. **Loading:** Place load corresponding to confinement pressure (100, 200, 300 kPa).

Step 9. **Consolidation:** Lower platform to initiate soil consolidation.

Step 10. **Testing:** Start machine at 1.5 mm/s, record dynamometric readings.

Step 11. **Failure:** Test continues until peak stress is surpassed and residual strength is reached.

4. EXPERIMENTAL RESULTS

The tests were conducted at three normal stresses: 100, 200, and 300 kPa. The detailed data logs for each test are provided in **Appendix A**.

4.1 Summary of Results

The following table summarizes the Peak and Permanent (residual) shear stresses observed.

Table 1: Summary of Shear Strength Values

Normal Stress σ_n (kPa)	Peak Shear Stress τ_{peak} (kPa)	Permanent Shear Stress τ_{perm} (kPa)
100	96.56	78.09
200	170.43	157.47
300	232.31	196.24

5. CALCULATIONS

SAMPLE CALCULATION (Test 1)

Taking the peak failure point from Test 1 where $\sigma_{nominal} = 100$ kPa.

1. Corrected Area (A_{corr}): The contact area decreases as the box shears. With horizontal displacement $\Delta L = 3.75$ mm:

$A_{corr} = L \times (L - \Delta L) = 60 \text{ mm} \times (60 - 3.75) \text{ mm} = 3375 \text{ mm}^2$

2. Normal Stress (σ_n):

$\sigma_n = \frac{N}{A} = \frac{0.36 \text{ kN}}{3600 \times 10^{-6} \text{ m}^2} = 100 \text{ kPa}$

3. Shear Stress (τ): The force ring reading indicated $F = 0.326$ kN at peak.

$\tau = \frac{F}{A_{corr}} = \frac{0.326 \text{ kN}}{3375 \times 10^{-6} \text{ m}^2} = 96.59 \text{ kPa}$

6.1 Graph 1: Stress-Strain Behavior

This graph shows the shear stress evolution. The curve exhibits a distinct peak followed by a reduction to permanent strength, typical of dense sand.

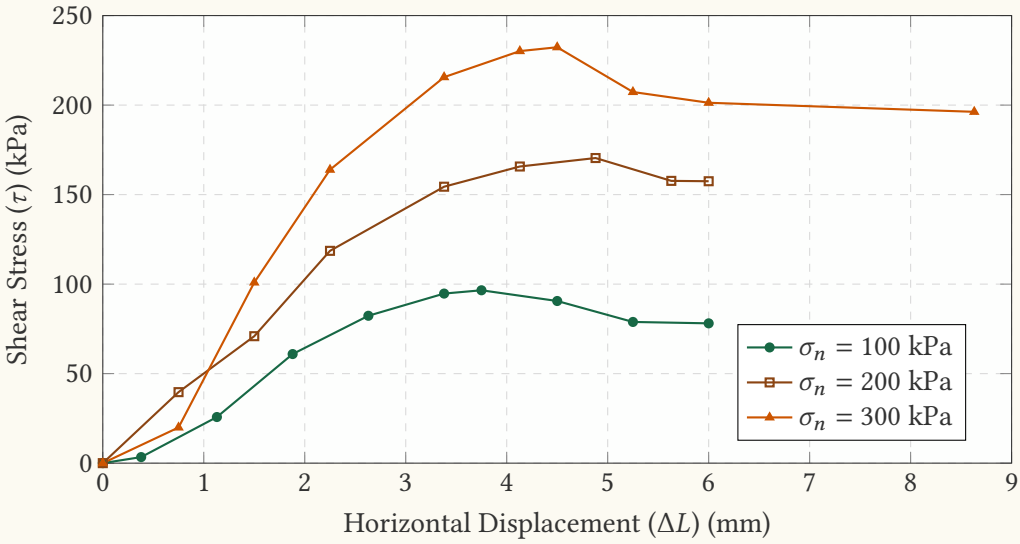

Figure 1: Shear Stress vs. Horizontal Displacement

6.2 Graph 2: Dilatancy (Vertical vs. Horizontal Displacement)

The curves show initial compression (negative) followed by significant dilation (positive), confirming the sample is dense.

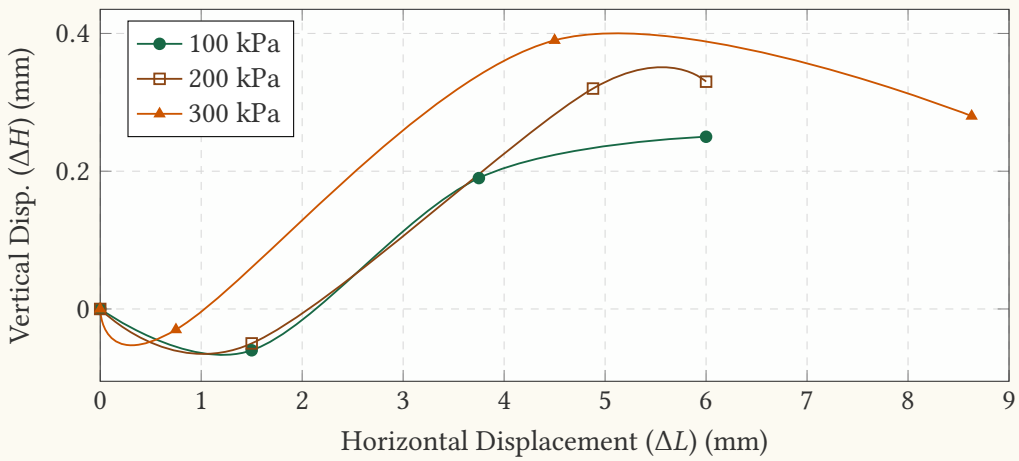

Figure 2: Dilatancy Curves (Positive = Expansion/Dilation)

6.3 Graph 3: Failure Envelopes (Mohr-Coulomb)

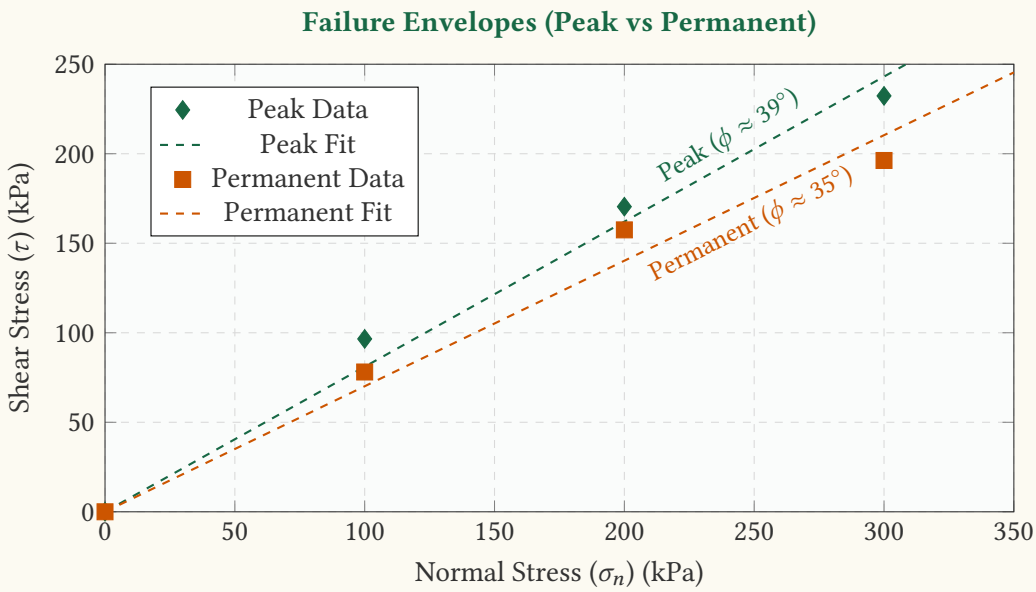

Figure 3: Mohr-Coulomb Failure Envelopes

7. DISCUSSION OF RESULTS

7.1 Interpretation of Curves

The stress-strain curves (Graph 1) clearly indicate the behavior of **dense sand**. We observe a distinct peak stress at relatively small strains (approx. 6-8% horizontal strain), followed by a "softening" behavior where the stress drops.

Graph 2 (Dilatancy) confirms this state. The initial negative displacement indicates slight compression as particles rearrange, followed by a strong positive vertical displacement. This **dilation** occurs because the densely packed grains must roll over each other to shear, causing an increase in sample volume.

7.2 Sources of Error

Despite the high correlation coefficient, potential sources of error include:

- **Machine Compliance:** Elastic deformation of the shear box apparatus itself.
- **Area Correction:** The simplified area correction assumes a perfect rectangle, whereas the actual failure plane is complex.
- **Loading Rate:** If the rate (1.5 mm/min) is too fast, pore water pressure might generate (though less critical in dry sand).

Calculated Shear Strength Parameters:

$c = 0$ **kPa** $\phi_{peak} \approx 39.02°$ $\phi_{perm} \approx 35.04°$

The direct shear test on the **dense sand** sample yielded the following conclusions:

- ✓ **Strength Behavior:** The material exhibits a clear peak strength followed by softening to a permanent state.
- ✓ **Friction Angle:** A high peak friction angle of 39.02° confirms the high density of the sample ($I_d = 0.9$). The permanent angle is lower (35.04°) as the sand grains rearrange and dilate.
- ✓ **Cohesion:** The regression analysis passes through the origin, confirming that the cohesion is effectively **zero**, which is theoretically correct for clean sand.

The tables below display the detailed data recorded during the test.
Legend: Peak Strength Permanent Strength

A.1 Test 1: $\sigma_n = 100$ kPa

Table A1: Data Log - 100 kPa

V.Disp (mm)	H.Disp (mm)	Reading (mm)	Force (kN)	ϵ_h (%)	ϵ_v (%)	Area (mm^2)	Stress (kPa)
0.00	0.00	0	0.00	0.00	0.00	3600.0	0.00
-0.02	0.38	6	0.012	0.63	-0.06	3577.5	3.38
-0.03	0.75	8	0.016	1.25	-0.09	3555.0	4.53
-0.05	1.13	45	0.091	1.88	-0.15	3532.5	25.76
-0.06	1.50	75	0.152	2.50	-0.18	3510.0	43.25
-0.04	1.88	105	0.213	3.13	-0.12	3487.5	60.93
-0.01	2.25	125	0.253	3.75	-0.03	3465.0	73.02
0.04	2.63	140	0.283	4.38	0.12	3442.5	82.32
0.08	3.00	151	0.306	5.00	0.24	3420.0	89.36
0.14	3.38	159	0.322	5.63	0.41	3397.5	94.72
0.19	**3.75**	**161**	**0.326**	**6.25**	**0.56**	**3375.0**	**96.56**
0.24	4.13	158	0.320	6.88	0.71	3352.5	95.39
0.27	4.50	149	0.302	7.50	0.79	3330.0	90.57
0.28	4.88	138	0.279	8.13	0.82	3307.5	84.44
0.28	5.25	128	0.259	8.75	0.82	3285.0	78.87
0.26	5.63	127	0.257	9.38	0.76	3262.5	78.77
0.25	6.00	125	0.253	10.00	0.74	3240.0	78.09

A.2 Test 2: $\sigma_n = 200$ kPa

Table A2: Data Log - 200 kPa

V.Disp	H.Disp	Reading	Force	ϵ_h	ϵ_v	Area	Stress
0.00	0.00	0	0.00	0.00	0.00	3600.0	0.00
-0.02	0.38	6	0.01	0.63	-0.06	3577.5	3.38
-0.03	0.75	7	0.14	1.25	-0.09	3555.0	39.66
-0.04	1.13	60	0.12	1.88	-0.12	3532.5	34.37
-0.05	1.50	123	0.25	2.50	-0.15	3510.0	70.91
-0.05	1.88	173	0.35	3.13	-0.15	3487.5	100.42
-0.04	2.25	203	0.41	3.75	-0.12	3465.0	118.59
0.01	2.63	230	0.47	4.38	0.03	3442.5	135.22
0.06	3.00	245	0.50	5.00	0.18	3420.0	145.00
0.12	3.38	259	0.52	5.63	0.35	3397.5	154.41
0.18	3.75	269	0.55	6.25	0.53	3375.0	161.54
0.24	4.13	274	0.56	6.88	0.71	3352.5	165.70
0.29	4.50	275	0.56	7.50	0.85	3330.0	167.45
0.32	**4.88**	**278**	**0.56**	**8.13**	**0.94**	**3307.5**	**170.43**
0.33	5.25	260	0.53	8.75	0.97	3285.0	160.33
0.33	5.63	254	0.51	9.38	0.97	3262.5	157.67
0.33	6.00	252	0.51	10.00	0.97	3240.0	157.47

A.3 Test 3: $\sigma_n = 300$ kPa

Table A3: Data Log - 300 kPa

V.Disp	H.Disp	Reading	Force	ϵ_h	ϵ_v	Area	Stress
0.00	0.00	0	0.00	0.00	0.00	3600.0	0.00
-0.02	0.38	35	0.02	0.63	-0.06	3577.5	5.59
-0.03	0.75	110	0.07	1.25	-0.09	3555.0	19.92
-0.03	1.13	175	0.22	1.88	-0.09	3532.5	63.01
-0.02	1.50	235	0.35	2.50	-0.06	3510.0	100.91
0.01	1.88	280	0.48	3.13	0.03	3487.5	136.40
0.05	2.25	335	0.57	3.75	0.15	3465.0	163.90
0.10	2.63	345	0.64	4.38	0.29	3442.5	185.88
0.16	3.00	360	0.70	5.00	0.47	3420.0	205.15
0.23	3.38	370	0.73	5.63	0.68	3397.5	215.60
0.28	3.75	379	0.75	6.25	0.82	3375.0	223.11
0.34	4.13	380	0.77	6.88	1.00	3352.5	230.16
0.39	**4.50**	**365**	**0.77**	**7.50**	**1.15**	**3330.0**	**232.31**
0.41	4.88	335	0.74	8.13	1.21	3307.5	224.55
0.40	5.25	323	0.68	8.75	1.18	3285.0	207.31
0.39	5.63	321	0.66	9.38	1.15	3262.5	201.16
0.38	6.00	318	0.65	10.00	1.12	3240.0	201.30
0.36	6.38	316	0.65	10.63	1.06	3217.5	200.78
0.35	6.75	314	0.64	11.25	1.03	3195.0	200.91
0.34	7.13	310	0.64	11.88	1.00	3172.5	201.04
0.32	7.50	303	0.63	12.50	0.94	3150.0	199.87
0.31	7.88	301	0.62	13.13	0.91	3127.5	196.71
0.29	8.25	298	0.61	13.75	0.85	3105.0	196.81
0.28	8.63	295	0.61	14.38	0.82	3082.5	196.24

"Geotechnical engineering is the art of building on the uncertain with certainty."